

MTH 732 Topology - Homework 5

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Assignment 1. Find all connected covering spaces of $\mathbb{RP}^2 \vee \mathbb{RP}^2$.

Proof. First, we have the universal cover, which is an infinite strip of spheres. For instance, you could model in $\mathbb{R}^3$ with spheres of radius 0.5, each centered at $(n, 0, 0)$ where n is an integer, so that there is a wedge point at each $(x, 0, 0)$ where $x - 0.5 \in \mathbb{Z}$. Call this universal cover E . Every other connected covering space will be a chain of n spheres, formed by the quotient $E/((x, 0, 0) \sim (x + nz, 0, 0))$. $\square$

Assignment 2. Find a space E for which there exists a 3-sheeted covering map $p : E \rightarrow T^2$ of the torus T^2 . Describe the mapping p .

Proof. Let E be $[0, 1] \times [0, 3]$, where $(x, 0) \sim (x, 3) \forall x \in [0, 1]$ and $(0, y) \sim (1, y) \forall y \in [0, 3]$. We will realize T^2 as the usual identification space on $\mathbb{R}^2/\mathbb{Z}^2$. The covering map $p : E \rightarrow T^2$ is then described as

$$p([(x, y)]_E) = ([x, y]_{T^2}).$$

We see that for any point $([x, y]_{T^2}) \in \mathbb{R}^2/\mathbb{Z}^2$, we have

$$p^{-1}([(x, y)]_{T^2}) = \{[(x, y)], [(x, y + 1)], [(x, y + 2)]\}.$$

For any $[(x, y)]_{T^2} \in T^2$, then, there is a neighborhood U such that $p^{-1}(U)$ is the union of three disjoint open neighborhoods of elements of $p^{-1}([(x, y)]_{T^2})$, each of which is homeomorphic to U . Thus, p is a covering map. $\square$

Assignment 3. Let X and Y be path-connected, locally path-connected, and semilocally simply connected spaces. Let $\tilde{X}$ and $\tilde{Y}$ be simply connected covers of X and Y , respectively. If X is homotopy equivalent to Y , show that $\tilde{X}$ is homotopy equivalent to $\tilde{Y}$.

Proof. Let $f : X \rightarrow Y$ be a homotopy equivalence and let $g : Y \rightarrow X$ be a homotopy inverse of f . Also, let $p_X : \tilde{X} \rightarrow X$ and $p_Y : \tilde{Y} \rightarrow Y$ be the covering maps. Then $f \circ p_X : \tilde{X} \rightarrow Y$ lifts (since $\pi_1(\tilde{X})$ is trivial and thus its image under $(f \circ p_X)_*$ is trivial) to a map $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$. Similarly, $g \circ p_Y$ lifts to a map $\tilde{g} : \tilde{Y} \rightarrow \tilde{X}$.

Now since $f \circ g \simeq \text{id}_Y$, $f \circ g \circ p_Y \simeq p_Y$. This homotopy can then be lifted along p_Y to get $\tilde{f} \circ \tilde{g} \sim \text{id}_{\tilde{Y}}$. Similarly, we get $\tilde{g} \circ \tilde{f} \sim \text{id}_{\tilde{X}}$. Thus $\tilde{X}$ and $\tilde{Y}$ are homotopy equivalent. $\square$

Assignment 4. Show that a torus cannot cover a sphere, and vice-versa.

Proof. Let $p : T^2 \rightarrow S^2$ be a covering map. Then p induces an injection from $\pi_1(T^2)$ to $\pi_1(S^2)$, which is impossible since $\pi_1(T^2)$ is non-trivial and $\pi_1(S^2)$ is trivial. In the other direction, if S^2 covers T^2 , then since S^2 is simply connected, it would be a universal cover, and thus it would be homeomorphic to the already-known universal cover $\mathbb{R}^2$. Since S^2 is not homeomorphic to $\mathbb{R}^2$, we are done. $\square$

Assignment 5. Describe a covering map $p : M_3 \rightarrow M_2$, where M_g is the connected, compact surface of genus g .

Proof. Not sure what to do here. $\square$